What is Cisco Packet Tracer?

Cisco Packet Tracer is a **network simulation software** developed by **Cisco Systems**. It allows us to **design, configure, and test computer networks** without using real networking devices.

Why do we use Cisco Packet Tracer?

Buying routers, switches, and cables for practice is expensive. Packet Tracer allows us to:

- Create computer networks virtually.
- Connect devices using different cables.
- Configure IP addresses.
- Configure routers and switches.
- Test network connectivity using commands like **ping**.

Real-Life Example

Suppose a company wants to connect **20 computers** through switches and routers.

Instead of buying all the equipment first, the network engineer:

1. Creates the network in Cisco Packet Tracer.
2. Tests whether all devices can communicate.
3. Fixes any problems.
4. Then implements the same network in the real world.

❖ Download and install cisco Packet tracer:

Step 1: Visit Cisco's Official Website

1. **Go to:** https://www.netacad.com
2. **Click "Log In" or "Sign Up" to create a Cisco Networking Academy (NetAcad) account (it's free).**

Step 2: Enroll in "Introduction to Packet Tracer" Course

1. After login, search for **"Introduction to Packet Tracer"**.
2. Enroll in the course (it's free and gives access to download links).

Step 3: Download Packet Tracer

1. Once enrolled, click on the **Resources** or **Download Packet Tracer** link.
2. Select your OS → **Windows 64-bit** or **Windows 32-bit**.
3. The file will download as a .exe installer (e.g., PacketTracer8.2.2.exe).

Step 4: Install Packet Tracer

1. Double-click the downloaded installer.
2. Click **Next** → accept the license agreement → **Next** → choose installation path → **Install**.
3. Wait for installation to complete.
4. Click **Finish**.

Step 5: Launch Packet Tracer

1. Open **Cisco Packet Tracer** from the Start menu or Desktop.
2. When asked to log in, use your **NetAcad account credentials**.
3. Once logged in, you'll see the Packet Tracer workspace — ready to start creating networks!

Hub

What is a Hub?

- A **Hub** is a **basic networking device** used to connect **multiple computers or network devices** in a **Local Area Network (LAN)**.
- It operates at the **Physical Layer (Layer 1)** of the **OSI Model**.
- A hub has **multiple ports**.
- It does **not** examine or filter data. When it receives a data packet on one port, it simply **copies (broadcasts)** the packet to **all other ports**.
- Every connected device receives the packet, but only the intended destination processes it; the others ignore it



Cable Used

- A **PC** and a **Hub** are different devices, so they are connected using an **Ethernet Straight-Through Cable**.

Example

Suppose **PC1**, **PC2**, and **PC3** are connected to a hub.

- If **PC1** sends data to **PC2**, the hub sends the same data to **PC2 and PC3**.
- **PC2** accepts the data because it is the intended receiver.
- **PC3** receives the data but ignores it because it is not the destination.

Hub Pros and cons:

Pros:

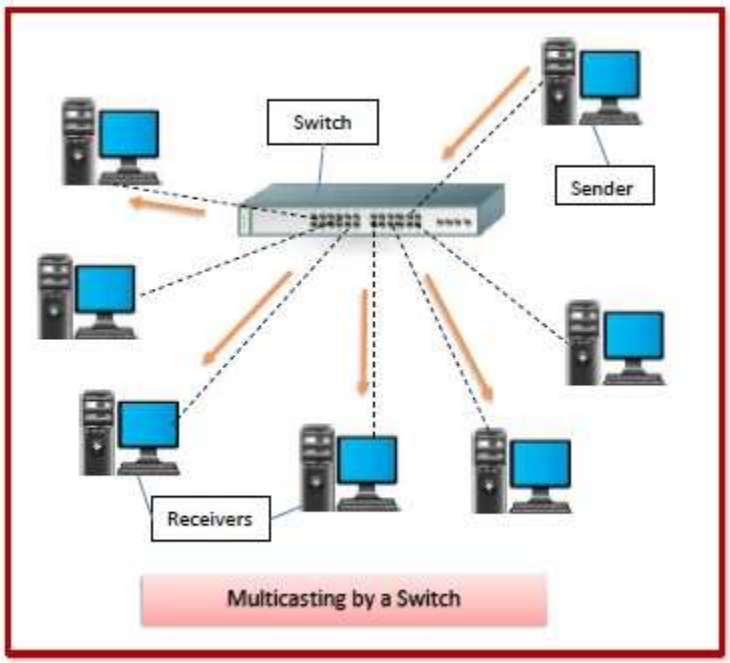
- Cheaper than switches.
- Works good for smaller network.

Cons:

- Issues with broadcast.
- No memory.

What is a Switch?

- A **Switch** is a networking device used to connect **multiple computers or devices** in a **Local Area Network (LAN)**.
- It operates at the **Data Link Layer (Layer 2)** of the **OSI Model**.
- A switch is **intelligent** because it sends data **only to the destination device**, not to all devices.
- It has **memory** to store a **MAC Address table**, which helps it forward data to the correct device.



Step 1: Open Cisco Packet Tracer

Start the software.

Step 2: Add Devices

1. From the **Device-Type Selection box**:
   - Click **End Devices** → drag **6 PCs** to the workspace.

- Click **Switches** → drag a **2960 Switch** (for example: "Switch0").

Step 3: Connect Devices

1. Click the **Lightning Bolt (Connections)** tool.
2. Choose **Copper Straight-Through** cable.
3. Connect:
   - PC0 → Switch Port **FastEthernet0/1**
   - PC1 → Switch Port **FastEthernet0/2**
   - PC2 → Switch Port **FastEthernet0/3**
   - PC3 → Switch Port **FastEthernet0/4**
   - PC4 → Switch Port **FastEthernet0/5**
   - PC5 → Switch Port **FastEthernet0/6**

Step 4: Assign IP Addresses

Click each PC → **Desktop → IP Configuration** and assign:

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- PC3→ 192.168.1.4
- PC4 → 192.168.1.5
- PC5 → 192.168.1.6
  (Subnet Mask: 255.255.255.0)

Step 5: Test the Network

1. Click **PC0 → Desktop → Command Prompt**.
2. ping 192.168.1.2

   You'll get a **Reply** from PC1
3. ping 192.168.1.3

Step 6: How to see the MAC Address.

1. Click on Switch
2. go to the CLI Mode, Press Enter,
3. Type EN

4. Type show mac -address / #show mac-address-table

Router

- A **Router** is a **network device** that connects **different networks** and forwards data from one network to another.
- It operates at the **Network Layer (Layer 3)** of the **OSI Model**.
- It determines the **best path** for data to reach its destination.
- A **Switch** connects devices within the **same LAN**, whereas a **Router** connects **different networks** (e.g., LAN to LAN or LAN to the Internet).

Step 1: Open Cisco Packet Tracer

- Start the software.

Step 2: Add Devices

- Add **1 Router** (e.g., `Router0` → model `1941` or `2911`).
- Add **2 Switches** (optional for neat design).
- Add **3 PCs** (PC0 , PC1,PC2) connect with one switch.
- Add **3 PCs**(PC3, PC4, PC5) connect with second switch.

Step 3: Assign IP Addresses

On PCs:

- **PC0** → IP: 10.10.10.1 | Subnet: 255.255.255.0 | Gateway: 10.10.10.4
- **PC1** → IP: 10.10.10.2 | Subnet: 255.255.255.0 | Gateway: 10.10.10.4
- **PC2** → IP: 10.10.10.3| Subnet: 255.255.255.0 | Gateway: 10.10.10.4
- **PC3** → IP: 192.168.1.1 | Subnet: 255.255.255.0|Gateway: 192.168.1.4
- **PC4** → IP: 192.168.1.2 | Subnet: 255.255.255.0|Gateway: 192.168.1.4
- **PC5** → IP: 192.168.1.3 | Subnet: 255.255.255.0 |Gateway: 192.168.1.4

Step 4: Connect Devices

Use **Copper Straight-Through** cables:

- PC0 → Switch0 → Router (GigabitEthernet0/0)
- PC1 → Switch1 → Router (GigabitEthernet0/1)

Step 5: Configure the Router

- Click the Router → go to Config

- Go to the GigabitEthernet0/0 → Set IP Address: 10.10.10.4 →Port status→on
- Go to the GigabitEthernet0/1 → Set IP Address: 192.168.1.4 →Port status→on

Step 6: Test the Network

1. **Click PC1 (on any pc of 1st lan network)→ Desktop → Command Prompt**
2. **ping 192.168.1.2(to communicate with any other device or second lan network)**

Topologies:

1. Star Topology in Cisco : A Star Topology is a network design where all devices (computers, printers, etc.) are connected to a central device, such as a switch or hub.

Step to Implement star topology:

- Devices Needed:
  - 1 switch or Hub (Centre Device)
  - Multiple PCs or End Device (usually 4 to 5 for Demonstration)
- Implementation:
  - Drag and drop switch from the device Panel.
  - Place PCs on the workspace.
  - Connect each PC to the Switch using Copper Straight Through cables(Click on the each device, "FastEhternet" and then choose the correct port to connect).
  - Config IP address on each PC:
    - Go to each PC →Desktop→**IP configuration.**
    - Assign unique Ips to each PC(For example 192.168.1.1, 192.168.1.2, etc..)
  - Test Connectivity using the Ping command.

2. Bus Topology

Bus Topology: A Bus Topology is a network topology in which all devices are connected to a single backbone cable. Since Cisco Packet Tracer does not provide a backbone cable, we implement it using multiple switches connected in a linear (bus-like) structure.

Devices Needed:

- 4 Switches
- 4 PCs

Implementation:

1. Open **Cisco Packet Tracer**.
2. Drag and drop **4 Switches** onto the workspace.
3. Arrange the switches in a **straight line**.
4. Connect the switches to each other using **Copper Cross-Over** cable.
    - Switch0 ↔ Switch1
    - Switch1 ↔ Switch2
    - Switch2 ↔ Switch3
5. Add **4 PCs**.
6. Connect each PC to a switch using **Copper Straight-Through** cable.
    - PC0 → Switch0
    - PC1 → Switch1
    - PC2 → Switch2
    - PC3 → Switch3
7. Configure IP addresses on all PCs:
    - Go to **PC → Desktop → IP Configuration**.
    - Assign IP addresses:
        - PC0 → **192.168.1.1**
        - PC1 → **192.168.1.2**
        - PC2 → **192.168.1.3**
        - PC3 → **192.168.1.4**
    - Subnet Mask: **255.255.255.0**
8. Test connectivity using the **Ping** command between any two PCs.

3. Ring Topology

A **Ring Topology** is a network topology in which each device is connected to **two neighboring devices**, forming a **closed loop (ring)**. Data travels around the ring in one or both directions until it reaches the destination. **Data travels in a circular path** through each device on the ring.

Devices Needed

- 4 PCs
- 4 Switches

Implementation

1. Open **Cisco Packet Tracer**.
2. Drag and drop **4 PCs** and **4 Switches** into the workspace.
3. Connect each **PC to its Switch** using a **Copper Straight-Through** cable.
4. Connect the **Switches in a ring** using **Copper Cross-Over** cables.
5. Configure IP addresses on all PCs (same subnet).
6. Test connectivity using the **ping** command.

Go to each PC →Desktop→IP Configuration.

Assign Unique Ips to each PC within the same subnet.

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- PC3 → 192.168.1.4
- Subnet Mask → 255.255.255.0

Note:

Cisco Packet Tracer does not provide a backbone cable for Bus topology or a real token for ring implementation. Therefore, both are demonstrated using switches and cables. The physical setup may look similar, but the concept and data flow are different.

4. Mesh Topology: In a Mesh Topology, every device (node) is directly connected to every other device in the network.

Devices Needed:

- Multiple PCs(at Least 4 Devices to demonstrate full mesh)
- Switch(atleast 4)

Implementation:

1. Open Cisco Packet Tracer.
2. Drag and drop Pcs and Switch into the workspace.
3. Connect Each switch to every switch using Copper Cross-Over cables.
4. Connect each PC to every other Switch using Copper straight- Through Cable
5. Assign Unique IP Address to each PC within the same subnet.
6. Test the connectivity using the **ping** command

5. Hybrid Topology:

Steps to Implement Hybrid Topology:

- Devices Needed:
  - 8PCs and 8 Switches
- Implementation:

1. Open Cisco Packet Tracer.

2. Drag and Drop several PCs into the workspace.

3. First make Bus Topology. And connect Pcs and switch as a bus topology.

4. In bus topology last Switch connect as a star topology so connect 3 PCs.

5. Assign IP addresses to each PC:

- Go to each PC →Desktop→IP Configuration.
- Assign Unique Ips to each PC within the same subnet.

6. Last make Ring Topology.

7. Test the connectivity by using Ping.

